

Momentum + Options Agents

AGENTIC HACKATHON

Alpaca Paper Trading

One-page summary: AI logic, risk gates, and the Alpaca infrastructure implementation behind an autonomous, six-agent momentum + options trading system.

1 · AI / AGENT LOGIC

A deterministic six-agent pipeline. An LLM can veto, but never generates or sizes a trade – kept out of the signal path so the strategy stays backtestable.

Market Data

Scores every liquid stock on vol-adjusted momentum: $0.45 \times (12\text{-}1\text{mo return}) + 0.25 \times (3\text{mo}) + 0.20 \times \text{trend} + 0.10 \times (\text{rel. strength vs SPY})$, each divided by 60-day realised vol. Reads real SEC EDGAR XBRL filings for post-earnings drift (SUE score + volume-confirmed gap must both agree).

Strategy

Rules only, no risk checks. Proposes: CORE (top-6 momentum, equal-weight, cash if SPY < 200-day SMA), PEAD (5%/event on a confirmed earnings beat), SPREAD (20-delta put credit spread on a name already held), SPREAD_EXIT (close at 50% profit or ≤ 5 DTE, unconditionally).

Judgment (AI, optional)

Can flag a name as risky. Cannot invent or size a trade. Abstains without an API key – the deterministic pipeline runs unchanged.

2 · RISK GATES

Every gate is a runtime check that *raises*, not a convention that could silently drift.

4 paper-trading gates – all required

PAPER_TRADING=true · paper API endpoint · account number prefix 'PA' · SHA-256 of the account number in a tracked source *allowlist* (not .env) – widening it is a reviewable code change.

Risk agent – hard veto authority

Position limits (max 10, 20% per name), overlay risk budget (15% max collateral, 3% per spread), and the *single-pot capital rule*: ``portfolio.py`` asserts $\text{cash} \geq 0$ and $\text{reserved collateral} \leq \text{cash}$ – money cannot be spent twice. An earlier accounting bug here inflated a 15-yr backtest CAGR by ~8pp before being caught.

Execution agent – the one-way valve

The *only* component that can write to Alpaca. Handed an unapproved decision, it *raises* rather than submits – structural, not a convention an agent could talk itself out of.

Pinned by tests

64 automated tests, including ones that specifically prove the capital invariants hold and that execution refuses an unapproved order.

3 · ALPACA INFRASTRUCTURE

APIs

Trading API (paper) for account/positions/orders; Market Data API for daily bars, quotes, and option chains (IEX feed).

Order types

Stock legs are simple market orders. Spreads are Alpaca's *multi-leg (mleg) limit orders* – the defined-risk order class options strategies require – priced from the resolved contracts' own live quotes, not the pricing model's.

Verification at runtime

``broker.py::verify_paper()`` calls Alpaca's account endpoint and checks the returned account number against the gates in §2 before a single order can be built.

Scheduling

No daemon. A Hermes cron job runs ``cli cycle --no-llm --execute`` 3x/weekday (9:45, 12:45, 3:45 ET) against the Alpaca paper endpoint; results post to Discord. A closed market silently rejects every proposal – safe by default.

LIVE, NOT JUST BACKTESTED

Sept 3, 10:33am ET: the risk-gated, Alpaca-verified pipeline above sold a real TD 115/105 put credit spread on paper capital (\$0.55/share credit, 2 contracts) – autonomously, on schedule.